

WASP-3b

R-1316

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_150808 · created 2026-07-31T19:04:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457242.837 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.042
Transit depth (Rp/Rs)^2	2.0 +/- 1.19 [%]
Orbital Inclination (inc)	80.01 +/- 2.84
Airmass coefficient 1 (a1)	3.4 +/- 1.95
Airmass coefficient 2 (a2)	-0.88 +/- 0.22
Scatter in the residuals of the lightcurve fit is	48.36 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	3.0
Transit Duration (day)	0.076 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.042 vs 0.10538 (deviation 0.85σ)
Duration fit vs published	0.076 d vs 0.1167 d (deviation 1.1σ)
Mid-time O–C	-5.1 min
Depth SNR	3.4
Residual scatter	48.36 %

Light curve

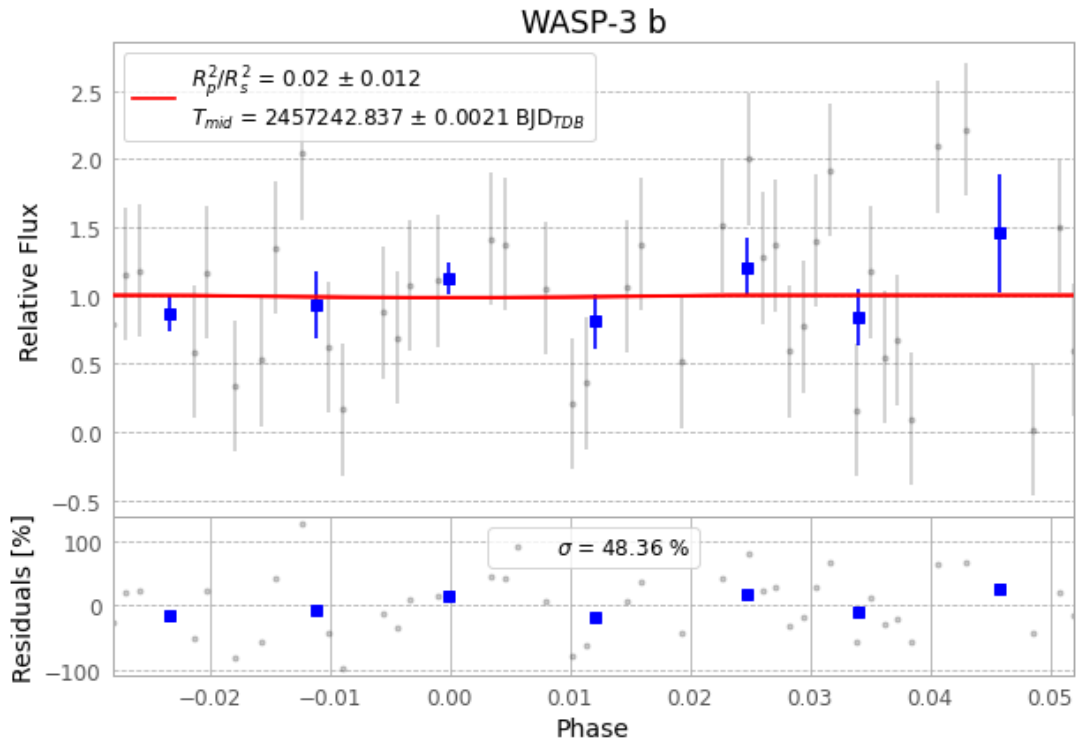


Figure 1 — FinalLightCurve_WASP-3 b_2015-08-07.png (SHA-256 27c6018382ed8446...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [201, 448], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2eb918d4ca1ee386...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), WASP-3150808064513.FITS → WASP-3150808101813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150808.log (806d44fd3641...), FinalLightCurve_WASP-3 b_2015-08-07.png (27c6018382ed...), FinalLightCurve_WASP-3 b_2015-08-07.csv (443ce7d77dbf...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:04Z.